higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

AUTOMOTIVE REPAIR AND MAINTENANCE
NQF LEVEL 2

22 NOVEMBER 2019

This marking guideline consists of 6 pages.

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QUESTION 1

1.1
1.1.1 E
1.1.2 B
1.1.3 A
1.1.4 F
1.1.5 C
(5 × 1) (5)

1.2
1.2.1 Safety goggles must be worn.
1.2.2 All the applicable PPE must be worn. (2 × 1) (2)

1.3
- Clean up immediately after spills.
- Place notices when the floor is wet.
- Workers must wear appropriate safety shoes. (Any 2 × 1) (2)

1.4
A – On/off switch
B – Spindle lock
C – Grinding/cutting disc
D – Safety guard
E – Handle (Any 3 × 1) (3)

1.5
- Keep yourself clean and as healthy as possible.
- Protect yourself from the spread of diseases.
- Wash your hands when you have finished working.
- Wash your hands before eating.
- Wash your hands after using the toilet.
- If you cut or graze your skin while working, make sure you clean and cover the wound to prevent infection. (Any 3 × 1) (3)

1.6
- Check that the cable is not open or damaged.
- Check the condition and size of the grinding stone.
- Check the adjustment of the tool-rest.
- Check that the eye shield and safety guards are in place.
- Check if it is in a working condition. (Any 2 × 1) (2)

1.7 Materials expand as the temperature increases. This means that the measurement of a cold part will be less than the measurement of the same part if it is warm or hot. (3)

1.8
1.8.1 A
1.8.2 B
1.8.3 B
1.8.4 C
1.8.5 D
(5 × 1) (5)

[25]

QUESTION 2

2.1 Batteries contain sulphuric acid that is very harmful for your clothing and it can cause severe burns to your skin. (2)

2.2 Polypropylene (1)

2.3 In series (1)

2.4
- It cleans the inside of the engine.
- It lubricates the moving parts.
- It dampens engine noises.
- It reduces friction.
- It prevents rust.
- It seals between piston rings and cylinder. (Any 3 × 1) (3)

2.5
2.5.1 False
2.5.2 True
2.5.3 True
2.5.4 False
2.5.5 True
2.5.6 False
2.5.7 True
(7 × 1) (7)

2.6
2.6.1 Anti-oxidant additive
2.6.2 Detergent additive
2.6.3 Viscosity modifier
(3 × 1) (3)

2.7
- Gear type
- Internal-gear type
- Eccentric rotor type (Any 2 × 1) (2)

2.8 Hydrodynamic lubrication occurs when the movement between the rotating parts is used to spread the lubricant around the bearing surface. (2)

2.9
2.9.1 Thermostat
2.9.2 Cooling system
2.9.3 Anti-freeze
2.9.4 Two-stroke
2.9.5 Aluminium
(5 × 1) (5)
[26]

QUESTION 3

3.1
- Lack of lubrication
- Overloading (excessive pressure/weight)
- Dirt entering the bearing
- Part failure
- Incorrect fitment (Any 3 × 1) (3)

3.2
- Rollers
- Outer race
- Inner race
- Bearing seal (Any 2 × 1) (2)

3.3
- It has more design options.
- It is lighter in weight.
- It has good thermal conductivity.
- It is corrosion resistant. (Any 3 × 1) (3)

3.4
- Location of the tyre manufacturer
- Who the manufacturer is
- Tyre construction type
- Date of tyre construction (Any 3 × 1) (3)

3.5
- Rotate the wheels on the vehicle at regular intervals.
- Regularly check and correct the pressure of the tyres.
- Monitor the tyre wear pattern and have the necessary adjustments made.
- Remove foreign objects from the tread.
- Avoid excessive speeding and hard braking. (Any 3 × 1) (3)

3.6 Hot water is pumped into the top tank of the radiator and it then passes through the radiator tubes. The wind that comes through the front grill and through the radiator cools down the fins of the radiator core. The fins in turn cool down the tubes and the tubes cool down the water. (3)

3.7 3.7.1
A – Camshaft sprockets
B – Piston
C – Crankshaft
D – Camshafts (4)

3.7.2
- With a cam belt/timing belt
- With a timing chain
- With meshed gears (Any 2 × 1) (2)

3.7.3
- It converts the up and down movement of the piston into a rotating motion.
- It supplies the rotating motion to the drive-line to turn the wheels. (2)

[25]

QUESTION 4

4.1 4.1.1 It must be adjusted upwards and to the left, where the horizontal and vertical lines cross.

4.1.2 It must be adjusted down to below the horizontal line and stay on the right side of the vertical line.

(2 × 2) (4)

4.2
- They use less power.
- They last longer.
- They use less space.
- They allow for better design choices. (Any 1 × 1) (1)

4.3
- Prepare the work area.
- Prepare the vehicle by fitting the necessary protection covers.
- Acquire all the tools needed for the task.
- Acquire the necessary documentation.
- Make sure all the parts are available. (Any 3 × 1) (3)

4.4
- Radiator coolant
- Windscreen washer water
- Battery water
- Brake fluid
- Power steering fluid
- Automatic transmission fluid (Any 3 × 1) (3)

4.5
- It extends the life span of the tool.
- It prevents you from starting on a new job with dirty tools.
- The tools will be available when you need them.
- No time will be wasted on looking for tools.
- You might injure yourself when you work with dirty tools. (Any 3 × 1) (3)

4.6 Engine oil would be thicker when the engine is cold. When draining cold oil, not all the dirty oil will drain out, as some oil will stick to the engine parts. Warm oil also drains out much quicker, saving you time. (3)

4.7
- Inside the vehicle
- Outside the vehicle
- Under the bonnet (3)

4.8
- It can cause soil pollution when dumped in the sand.
- It can cause water pollution when poured down the drain.
- It also causes air pollution when it is left open or when it is burnt.

(Any 2 × 1) (2)

4.9
- Spanners
- Sockets
- Hammer
- Screwdriver
- Ratchet

(Any 2 × 1) (2)

[24]

TOTAL: 100

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